

Sales Performance Analysis Project Report

Project Title: Sales Performance Analysis Dashboard

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Internship Role: Data Analyst

Tools & Technologies: Python, Power BI, Pandas, NumPy

Dataset: DQLab Store Sales Dataset

Time Period Covered: 2009 – 2012

Abstract:

- This project presents a comprehensive sales performance analysis conducted on four years of retail transaction data. The objective of the study is to transform raw, unstructured sales data into meaningful insights that support strategic decision-making. Using Python for data preprocessing and Power BI for visualization, the project demonstrates how analytics can uncover trends, improve operational efficiency, and identify revenue growth opportunities.

Introduction:

- In the modern business environment, data plays a critical role in shaping strategic and operational decisions. Organizations generate vast amounts of transactional data every day, but raw data in its original form is often unstructured, inconsistent, and difficult to interpret. Without proper analysis, valuable insights remain hidden within this data.
- Sales performance analysis is one of the most important applications of data analytics. It helps organizations understand how products, customers, and categories contribute to overall revenue. By studying historical sales data, businesses can identify growth patterns, seasonal trends, and areas where efficiency can be improved.
- This project focuses on analyzing historical sales data using Python and Power BI. Python is used to clean, preprocess, and transform the raw dataset into a reliable analytical format, while Power BI is used to design an interactive dashboard. The project demonstrates a complete analytics workflow, starting from raw data and ending with actionable business insights, without introducing any techniques or components beyond those actually implemented in the project.

Problem Statement

- The organization maintained sales records in a raw, pipe-separated format that was not suitable for direct analysis or reporting. The dataset contained several issues such as inconsistent date formats, unnecessary spaces in text fields, missing values, and incorrect data types for numeric columns. These problems made it difficult to calculate accurate metrics such as revenue and discounts.
- Additionally, decision-makers relied on static reports that did not allow filtering or deep exploration of data. As a result, identifying sales trends, high-value customers, and discount-heavy products required significant manual effort and time.
- The problem addressed by this project was to convert raw sales data into a clean, structured dataset and present it through an interactive dashboard that enables faster and more reliable decision-making.

Project Objectives

The primary objective of this project is to perform a comprehensive sales performance analysis using historical transaction data. To achieve this goal, the project focuses on the following specific objectives:

- To clean and preprocess raw sales data using Python
- To standardize date, numeric, and text fields for consistency
- To calculate net sales by accounting for discounts
- To analyze revenue trends over time
- To evaluate sales performance by product category
- To identify top revenue-generating customers
- To analyze discount patterns across products
- To design an interactive Power BI dashboard that reflects all findings

All objectives are strictly aligned with the actual tasks and analyses performed in the project.

Scope of the Project

- The scope of this project is confined to descriptive sales analysis based on historical data from the DQLab Store Sales Dataset covering the period from 2009 to 2012. The analysis includes revenue, order volume, quantity sold, customer contribution, category-wise performance, and discount behavior.
- The project does not include predictive modeling, forecasting, machine learning, or external data integration. All insights are derived solely from the provided dataset and visualized using Power BI. This well-defined scope ensures clarity, accuracy, and alignment with the actual implementation.

Dataset Description

- The dataset used for this project is the DQLab Store Sales Dataset, which contains transactional sales records for a retail business. Each row in the dataset represents a single order transaction.
- Key fields in the dataset include order ID, order date, customer name, product category, sales value, discount value, quantity sold, and order status. The dataset spans four years, providing sufficient historical depth to analyze long-term trends and seasonal patterns.
- This dataset serves as the foundation for all preprocessing, analysis, and visualization performed in the project.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	order_id	order_status	customer	order_date	order_quantity	sales	discount	discount_value	product_category	product_sub_category					
2															
3	3	Order Finished	Muhammed MacIntyre	2010-10-13	6	523080	0.04	20923	Office Supplies	Storage & Organization					
4	293	Order Finished	Barry French	2012-10-01	49	20246040	0.07	1417223	Office Supplies	Appliances					
5	483	Order Finished	Clay Rozendal	2011-07-10	30	9931519	0.08	794522	Technology	Telephones and Communication					
5	515	Order Finished	Carlos Soltero	2010-08-28	19	788540	0.08	63083	Office Supplies	Appliances					
7	613	Order Finished	Carl Jackson	2011-06-17	12	187080	0.03	5612	Office Supplies	Binders and Binder Accessories					
3	643	Order Finished	Monica Federle	2011-03-24	21	5563640	0.07	389455	Office Supplies	Storage & Organization					
9	678	Order Returned	Dorothy Badders	2010-02-26	44	456820	0.07	31977	Office Supplies	Paper					
0	807	Order Finished	Neola Schneider	2010-11-23	45	393700	0.01	3937	Office Supplies	Paper					
1	868	Order Finished	Carlos Daly	2012-06-08	32	1433680	0	0	Office Supplies	Appliances					
2	933	Order Finished	Claudia Miner	2012-08-04	15	161220	0.02	3224	Office Supplies	Binders and Binder Accessories					
3	995	Order Finished	Neola Schneider	2011-05-30	46	3630980	0.03	108929	Furniture	Office Furnishings					
4	998	Order Finished	Allen Rosenblatt	2009-11-25	16	496520	0.07	34756	Office Supplies	Envelopes					
5	1154	Order Finished	Sylvia Foulston	2012-02-14	44	8924460	0.04	356978	Furniture	Bookcases					
6	1344	Order Finished	Jim Radford	2012-04-15	15	1669808	0.06	100188	Technology	Telephones and Communication					
7	1412	Order Finished	Carlos Soltero	2010-03-12	13	118060	0.1	11806	Office Supplies	Labels					
8	1539	Order Finished	Carl Ludwig	2011-03-09	33	1023660	0.1	102366	Office Supplies	Binders and Binder Accessories					
9	1540	Order Finished	Don Miller	2012-08-04	30	161800	0.09	14562	Office Supplies	Pens & Art Supplies					
0	1702	Order Finished	Annie Cyprus	2011-05-06	23	134480	0.06	8069	Office Supplies	Pens & Art Supplies					
1	1761	Order Finished	Carl Ludwig	2010-12-23	25	24056460	0.01	240565	Technology	Copiers and Fax					
2	1792	Order Finished	Carlos Soltero	2010-11-08	28	740960	0.04	29638	Office Supplies	Storage & Organization					
3	2275	Order Finished	Grant Carroll	2012-10-21	49	556000	0.08	44480	Office Supplies	Pens & Art Supplies					
4	2277	Order Finished	Don Miller	2011-01-01	10	133080	0.01	1331	Technology	Computer Peripherals					

Data Challenges

- Several data quality challenges were identified during the initial exploration of the raw dataset. The data was stored in a pipe-separated format, which resulted in empty columns at the beginning and end of the table when loaded into Python.
- Text fields such as product categories and customer names contained leading and trailing spaces, which could cause duplication issues during analysis. Date values were stored as text, preventing time-based analysis. Numeric fields such as sales and discount values were also stored as text, increasing the risk of calculation errors.
- Addressing these challenges through preprocessing was essential to ensure accurate and reliable analysis.

Tools and Technologies

This project utilizes a combination of Python and Power BI to perform end-to-end sales performance analysis. Each tool was selected based on its strengths and relevance to the tasks involved in the project.

Python was used primarily for data preprocessing and transformation. Its ability to handle large datasets efficiently made it suitable for cleaning raw sales data, standardizing formats, and creating derived metrics such as Net Sales. The Pandas library enabled structured data manipulation, while NumPy supported numerical conversions and calculations.

Power BI was used for data visualization and reporting. After preprocessing, the cleaned dataset was imported into Power BI to create interactive dashboards. Power BI's visual elements such as cards, line charts, bar charts, treemaps, and slicers allowed insights to be presented in a clear and business-friendly manner.

The combined use of Python and Power BI ensured accuracy, scalability, and effective communication of insights.

Data Preprocessing Overview

Data preprocessing is a crucial step in the analytics lifecycle, as the quality of insights depends heavily on the quality of data. In this project, preprocessing was performed using Python with the help of Pandas and NumPy libraries.

The preprocessing phase involved reading the raw data file, assigning appropriate column names, removing unnecessary columns, cleaning text fields, standardizing data types, handling missing values, and creating derived metrics. Each step was carefully executed to ensure that the final dataset accurately represents real business transactions.

```
import pandas as pd
import numpy as np

# LOAD THE DATA

df_raw = pd.read_csv('raw_data.csv', sep='|', skiprows=2, header=None, engine='python')
```

Column Mapping and Structuring

The raw dataset was stored in a pipe-separated format, which resulted in unnecessary empty columns when loaded into Python. To address this, the dataset was carefully mapped into ten meaningful business columns that represent essential sales attributes.

Column mapping ensured that each field such as order ID, sales value, discount value, category, and customer name was correctly positioned and labeled. This step was crucial because incorrect or misaligned columns can lead to inaccurate analysis and misleading results.

By structuring the dataset properly, the foundation was laid for reliable preprocessing, accurate calculations, and seamless integration with Power BI.

```
# ORGANIZE COLUMNS

column_names = [
    'order_id', 'order_status', 'customer', 'order_date',
    'order_quantity', 'sales', 'discount', 'discount_value',
    'product_category', 'product_sub_category'
]
df = df_raw.iloc[:, 1:11]
df.columns = column_names
```

Data Cleaning Techniques

Data cleaning was performed to eliminate inconsistencies and errors present in the raw dataset. Text-based columns such as product category and customer name contained extra spaces, which could cause duplication issues during grouping and aggregation.

These issues were resolved by applying string manipulation techniques to remove leading and trailing spaces. Blank rows, particularly those without order IDs, were removed to ensure that order counts remained accurate.

This cleaning process improved data consistency and ensured that similar values were treated as a single entity during analysis.

```
# CLEAN TEXT & DATES

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = df[col].str.strip()

# format
df['order_date'] = pd.to_datetime(df['order_date'], errors='coerce')
```

Date and Numeric Standardization

Standardizing date and numeric fields was essential for performing accurate calculations and time-based analysis. Order dates were initially stored as text, which prevented chronological sorting and trend analysis.

Using Python, these date values were converted into proper datetime format. Numeric fields such as sales and discount values were also converted from text to numeric data types. Any invalid values were safely handled to prevent errors during analysis.

This standardization enabled reliable revenue calculations and effective visualization in Power BI.

Feature Engineering

Feature engineering in this project was limited to creating a single but critical business metric called Net Sales. Gross sales alone do not reflect the actual revenue earned by the business because discounts reduce the final amount received.

Net Sales was calculated by subtracting the discount value from the sales value for each transaction. This derived column provides a more realistic measure of revenue and was used consistently across all analyses and dashboard visuals.

No additional features or advanced transformations were introduced beyond what was actually implemented in the project.

```
# CLEAN NUMERICAL DATA
numeric_cols = ['order_id', 'order_quantity', 'sales', 'discount', 'discount_value']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

# Create Net Sales (Revenue after discounts)
df['net_sales'] = df['sales'] - df['discount_value']

# Remove any empty rows
df = df.dropna(subset=['order_id'])
```

Data Validation

After completing preprocessing and feature creation, data validation was performed to ensure accuracy and completeness. Summary checks such as reviewing row counts, inspecting sample records, and verifying calculated values were conducted.

A manual inspection of the first few rows of the cleaned dataset helped confirm that columns were correctly aligned and values appeared reasonable. This step reduced the risk of propagating errors into the dashboard.

Only after successful validation was the dataset exported and connected to Power BI for visualization.

Cleaned Dataset:

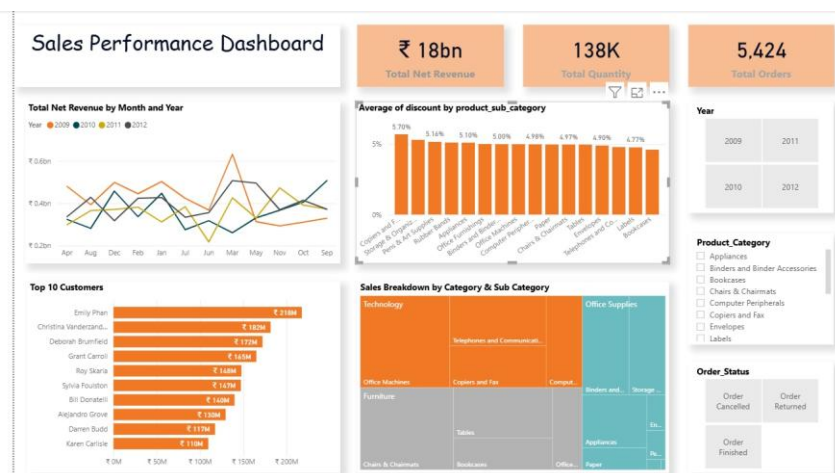
A	B	C	D	E	F	G	H	I	J	K
order_id	order_status	customer	order_date	order_quantity	sales	discount	discount_value	product_category	product_sub_category	net_sales
3	Order Finished	Muhammed MacIntyre	13-10-2010	6	523080	0.04	20923	Office Supplies	Storage & Organization	502157
293	Order Finished	Barry French	01-10-2012	49	20246040	0.07	1417223	Office Supplies	Appliances	18828817
483	Order Finished	Clay Rozendal	10-07-2011	30	9931519	0.08	794522	Technology	Telephones and Communication	9136997
515	Order Finished	Carlos Soltero	28-08-2010	19	788540	0.08	63083	Office Supplies	Appliances	725457
613	Order Finished	Carl Jackson	17-06-2011	12	187080	0.03	5612	Office Supplies	Binders and Binder Accessories	181468
643	Order Finished	Monica Federle	24-03-2011	21	5563640	0.07	389455	Office Supplies	Storage & Organization	5174185
678	Order Returned	Dorothy Badders	26-02-2010	44	456820	0.07	31977	Office Supplies	Paper	424843
807	Order Finished	Neola Schneider	23-11-2010	45	393700	0.01	3937	Office Supplies	Paper	389763
868	Order Finished	Carlos Daly	08-06-2012	32	1433680	0	0	Office Supplies	Appliances	1433680
933	Order Finished	Claudia Miner	04-08-2012	15	161220	0.02	3224	Office Supplies	Binders and Binder Accessories	157996
995	Order Finished	Neola Schneider	30-05-2011	46	3630980	0.03	108929	Furniture	Office Furnishings	3522051
998	Order Finished	Allen Rosenblatt	25-11-2009	16	496520	0.07	34756	Office Supplies	Envelopes	461764
1154	Order Finished	Sylvia Foulston	14-02-2012	44	8924460	0.04	356978	Furniture	Bookcases	8567482
1344	Order Finished	Jim Radford	15-04-2012	15	1669808	0.06	100188	Technology	Telephones and Communication	1569620
1412	Order Finished	Carlos Soltero	12-03-2010	13	118060	0.1	11806	Office Supplies	Labels	106254
1539	Order Finished	Carl Ludwig	09-03-2011	33	1023660	0.1	102366	Office Supplies	Binders and Binder Accessories	921294
1540	Order Finished	Don Miller	04-08-2012	30	161800	0.09	14562	Office Supplies	Pens & Art Supplies	147238
1702	Order Finished	Annie Cyprus	06-05-2011	23	134480	0.06	8069	Office Supplies	Pens & Art Supplies	126411
1761	Order Finished	Carl Ludwig	23-12-2010	25	24056460	0.01	240565	Technology	Copiers and Fax	23815895
1792	Order Finished	Carlos Soltero	08-11-2010	28	740960	0.04	29638	Office Supplies	Storage & Organization	711322
2275	Order Finished	Grant Carroll	21-10-2012	49	556000	0.08	44480	Office Supplies	Pens & Art Supplies	511520
2277	Order Finished	Don Miller	01-01-2011	10	133080	0.01	1331	Technology	Computer Peripherals	131749
2532	Order Finished	Alan Barnes	10-10-2011	30	564140	0.03	16924	Office Supplies	Labels	547216

Power BI Dashboard Design

The Power BI dashboard was designed with a focus on clarity, simplicity, and usability. The layout follows a top-down approach, where key performance indicators are displayed at the top, followed by detailed analytical visuals.

Visuals were arranged logically to guide users from high-level metrics to deeper insights. Consistent color usage and labeling were applied to ensure readability and reduce cognitive load.

The dashboard design allows decision-makers to quickly understand overall performance while also exploring specific aspects of the data.



Key Performance Indicators Analysis

Key Performance Indicators (KPIs) provide a concise summary of overall business performance. In this project, KPIs were created to display total net revenue, total number of orders, and total quantity sold.

These metrics were presented using card visuals and formatted in Indian Rupees where applicable. Placing KPIs at the top of the dashboard ensures that users can grasp the business status within seconds of viewing the report.

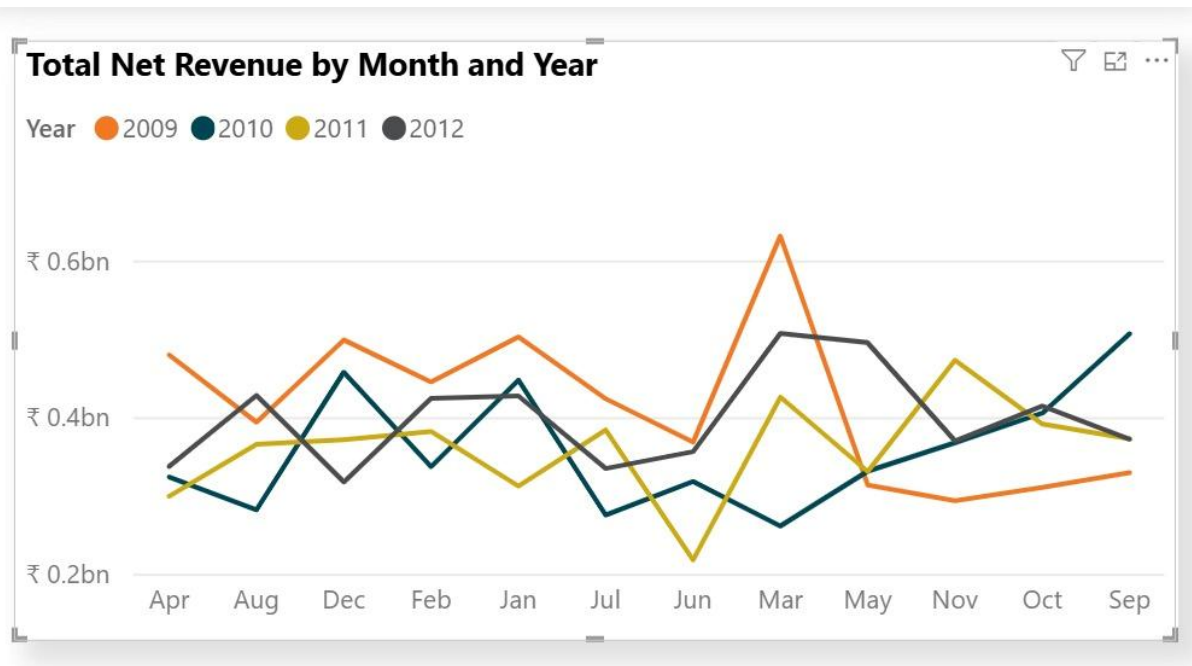
KPIs serve as a reference point for interpreting all other visuals in the dashboard.



Revenue Trend Analysis

Revenue trend analysis was conducted using a line chart in Power BI, where total net sales were plotted against order dates. This visualization allowed the identification of long-term growth patterns and seasonal variations over the four-year period.

The analysis revealed a recurring peak in sales during the month of March each year, indicating a consistent seasonal surge. Conversely, a noticeable dip in revenue was observed around the mid-year period, particularly in June. These trends provide valuable insights for planning inventory levels, staffing, and promotional activities.

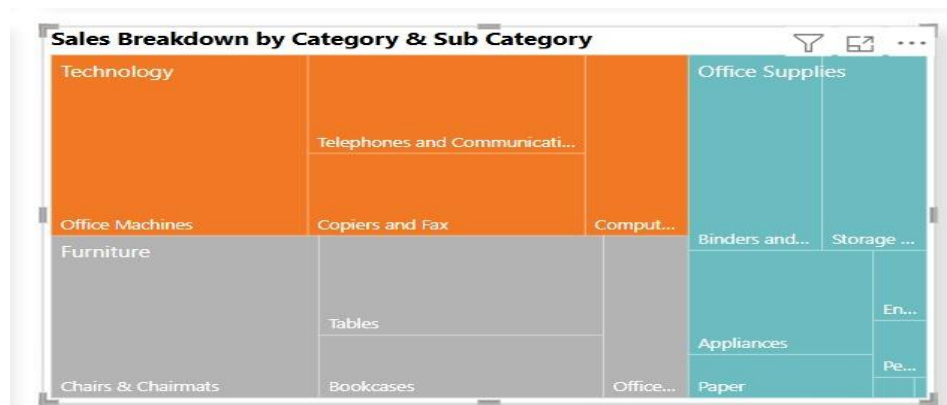


Category and Customer Analysis

Category-wise analysis was performed to understand how different product segments contribute to overall revenue. A treemap visualization was used to represent the proportion of net sales generated by each category.

Customer analysis focused on identifying the top revenue-generating customers. A bar chart was used to display the top ten customers based on net sales. This analysis highlights the importance of high-value customers and supports targeted account management strategies.

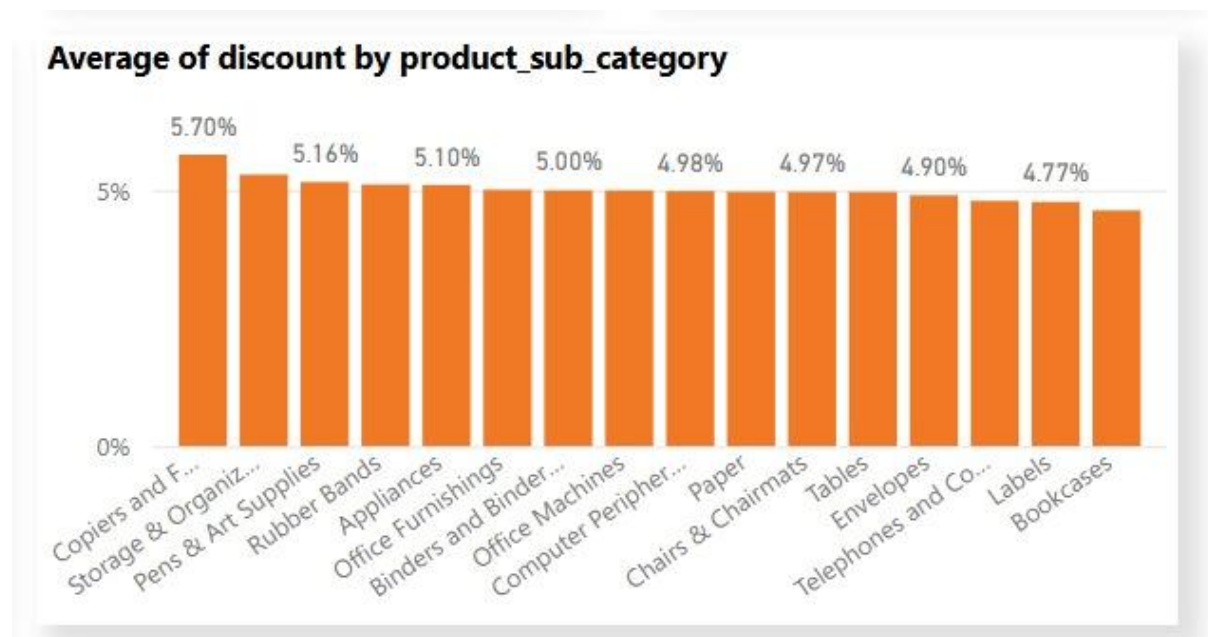
Together, category and customer analysis provide insights into where the business generates most of its value.



Discount Analysis

Discount analysis focused on understanding how pricing strategies impact revenue. Instead of analyzing total discounts, average discount percentages were calculated for each product category. This approach provides a clearer view of discount behavior across products.

The analysis showed that products such as Copiers and Fax machines receive significantly higher average discounts compared to other categories. This finding highlights potential areas of profit leakage and suggests that the existing discount strategy for these products should be reviewed.



Interactive Features

Interactive slicers were added to the dashboard to enhance user experience and analytical flexibility. Slicers were created for year, product category, and order status.

These slicers allow users to filter the entire dashboard dynamically with a single selection. This interactivity eliminates the need for multiple static reports and enables faster exploration of specific scenarios.

Interactive features significantly increase the practical value of the dashboard for business users.

The image displays three interactive slicers stacked vertically. The first slicer, titled 'Year', contains four buttons labeled '2009', '2011', '2010', and '2012'. The second slicer, titled 'Product_Category', features a list of categories with checkboxes: 'Appliances', 'Binders and Binder Accessories', 'Bookcases', 'Chairs & Chairmats', 'Computer Peripherals', 'Copiers and Fax', 'Envelopes', and 'Labels'. The third slicer, titled 'Order_Status', contains three buttons labeled 'Order Cancelled', 'Order Returned', and 'Order Finished'. Each slicer is enclosed in a light gray box with a subtle shadow.

Year	
2009	2011
2010	2012

Product_Category
☐ Appliances
☐ Binders and Binder Accessories
☐ Bookcases
☐ Chairs & Chairmats
☐ Computer Peripherals
☐ Copiers and Fax
☐ Envelopes
☐ Labels

Order_Status
Order Cancelled
Order Returned
Order Finished

Business Insights

The analysis generated several actionable business insights. Seasonal trends revealed consistent sales peaks during March, suggesting the need for higher inventory levels and staffing during this period.

The identification of discount-heavy products highlighted opportunities to improve profitability by revisiting pricing strategies. Additionally, the concentration of revenue among top customers emphasizes the importance of customer retention.

These insights are derived directly from the dashboard visuals and support informed decision-making.

Limitations of the Study

This study is limited to descriptive analysis of historical sales data. The insights are based solely on the available dataset and do not account for external factors such as market conditions or competition.

Additionally, the analysis reflects past performance and does not attempt to predict future outcomes. Despite these limitations, the findings remain valuable for understanding historical trends and operational patterns.

Future Enhancements

Future enhancements to this project could include extending the analysis to additional time periods if more data becomes available. Further breakdowns by region or product subcategories could also provide deeper insights.

Any future work should build upon the existing preprocessing and dashboard framework developed in this project.

Conclusion

This project demonstrates the effective use of Python and Power BI to transform raw sales data into meaningful insights. Through systematic preprocessing and clear visualization, complex data was converted into an accessible and interactive dashboard.

The analysis highlights key trends, identifies revenue drivers, and uncovers areas for improvement, such as discount management. Overall, the project showcases the practical application of data analytics in solving real-world business problems.

References

- <https://github.com/uchitesh8-web/Sales-Performance-Dashboard/tree/main>